Quadratic Equations

A quadratic equation in the variable x is an equation of the form:

$$ax^2 + bx + c = 0$$

where a, b, c are constant (at least constant with respect to x) and $a \neq 0$.

The 3 Methods of Solving Quadratic Equations

1) Factoring
2) Completing-the-Square
3) Quadratic Formula

$\left.\begin{array}{l}\text{2) Completing-the-Square} \\ \text{3) Quadratic Formula}\end{array}\right\} \to$ always work

Factoring:

This technique relies on the zero factor theorem:

If $a \cdot b = 0$, then $a = 0$ or $b = 0$
(or both)

e.g. $\underbrace{3x^2 = 10 - x}$

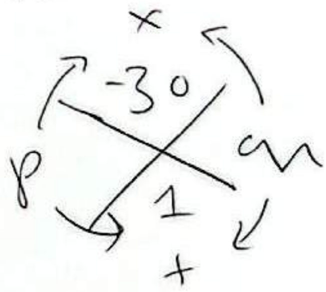
$$3x^2 + x - 10 = 0$$

STANDARD FORM

$$3x^2 - 5x + 6x - 10 = 0$$

$$x(3x-5) + 2(3x-5) = 0$$

a, b, c
$a = 3$, $b = 1$, $c = -10$



p	q
-1	30
-2	15
-3	10
-5	6

$$(x+2)(3x-5) = 0$$

$\underbrace{x+2}_{=0}$ OR $\underbrace{3x-5}_{=0}$

$x + 2 = 0$ or $3x - 5 = 0$

$\boxed{x = -2}$ OR $\boxed{x = \frac{5}{3}}$

g. $x^2+16=8x$

$$x^2-8x+16=0$$

x^2 $\quad 2\cdot x\cdot 4 \quad 4^2$

$$\sqrt{(x-4)^2}=\sqrt{0}$$

$$x-4=\pm 0$$

$$x-4=0$$

$$\boxed{x=4}$$

$$(x-4)^2=0$$

$$(x-4)(x-4)=0$$

$$x-4=0 \Rightarrow \boxed{x=4}$$

e.g. $x^2=25$

$$x^2-25=0$$

$x^2 \quad 5^2$

$$(x-5)(x+5)=0$$

$$x-5=0 \qquad x+5=0$$

$$\boxed{x=5} \qquad \boxed{x=-5}$$

ompleting-the-Square

We know how to solve equations like

e.g. $\sqrt{x^2} = \sqrt{5}$

$|x| = \sqrt{5}$

$x = \sqrt{5}$ or $-\sqrt{5}$

$\boxed{x = \pm\sqrt{5}}$ directly because of absolute value

&

e.g. $\sqrt{(x+3)^2} = \sqrt{5}$

$x + 3 = \pm\sqrt{5}$

$\boxed{x = -3 \pm \sqrt{5}}$

$x = -3 + \sqrt{5}$ OR $x = -3 - \sqrt{5}$

n we convert quadratic equations into a form where we may solve by just square-rooting both sides?

General Technique:

$$\underline{a}x^2 + \underline{b}x + c = 0$$

$a \neq 0$

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

"completing-the-square" step

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} + \frac{c}{a} = \frac{b^2}{4a^2}$$

$$\Rightarrow x^2 + 2x\left(\frac{b}{2a}\right) + \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = \frac{b^2}{4a^2}$$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} = \frac{b^2}{4a^2}$$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

...hat value of d makes the following expressions completed squares?

e.g. $x^2 - 3x + d \stackrel{?}{=} (x+c)^2$

$x^2 - 3x + d \stackrel{?}{=} x^2 + 2cx + c^2$.

$-3 = 2c$ → by comparing x-coeff.s

→ $c = \frac{-3}{2}$

$d = c^2$ → comparing constant terms

$\Rightarrow d = \left(\frac{-3}{2}\right)^2$

$\boxed{d = \frac{9}{4}}$

e.g. $x^2 + dx + 64$

$= (x+c)^2$

$x^2 + dx + 64 = x^2 + 2cx + c^2$

$d = 2c$

$64 = c^2$ → $c = \pm 8$

$d = 2c$ if $c = 8$

$\boxed{d = 16}$

if $c = -8$

$\boxed{d = -16}$

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g. 1. $x^2-5x+3=0$ solve by completing-the-square

$$x^2-5x+3=0$$

We want x^2-5x+d to be a completed square for some d.

$$x^2-5x+d=(x+c)^2$$
$$x^2-5x+d=x^2+2cx+c^2$$
$$-5=2c \Rightarrow c=-\frac{5}{2}$$
$$d=c^2 \Rightarrow d=\left(\frac{-5}{2}\right)^2=\frac{25}{4}$$

$$x^2-5x+3=0$$
$$\left[x^2-5x+\frac{25}{4}\right]+3=\frac{25}{4}$$
$$\left(x-\frac{5}{2}\right)^2+3=\frac{25}{4}$$
$$\left(x-\frac{5}{2}\right)^2=\frac{13}{4}$$

$$\sqrt{\left(x-\frac{5}{2}\right)^2}=\sqrt{\frac{13}{4}}$$

$$\Rightarrow x-\frac{5}{2}=\pm\sqrt{\frac{13}{4}}=\pm\frac{\sqrt{13}}{\sqrt{4}}=\pm\frac{\sqrt{13}}{2}$$

$$\Rightarrow x=\frac{5}{2}\pm\frac{\sqrt{13}}{2}$$

$$\Rightarrow \boxed{x=\frac{5\pm\sqrt{13}}{2}}$$

going back to our general completing-the-square form:

$$ax^2+bx+c=0 \implies \left(x+\frac{b}{2a}\right)^2 = \underbrace{\frac{b^2}{4a^2}-\frac{c}{a}}_{\text{LCD on right side}}$$

$$\implies \left(x+\frac{b}{2a}\right)^2 = \frac{b^2-4ac}{4a^2}$$

$$\implies x+\frac{b}{2a} = \pm\sqrt{\frac{b^2-4ac}{4a^2}}$$

$$\implies x+\frac{b}{2a} = \frac{\pm\sqrt{b^2-4ac}}{\sqrt{4a^2}} = \frac{\pm\sqrt{b^2-4ac}}{2a}$$

QUADRATIC FORMULA

$$\implies \boxed{x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}}$$

$b^2-4ac \rightarrow$ discriminant

Discriminant	Nature of roots
Positive	Two distinct real roots
Zero	One real root
Negative	Two distinct complex roots / no real roots

e.g. $9x^2 - 30x + 25$

$a = 9$

$b = -30$

$c = 25$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{30 \pm \sqrt{(-30)^2 - 4 \cdot 9 \cdot 25}}{2 \cdot 9}$$

$$= \frac{30 \pm \sqrt{0}}{18} = \frac{30}{18} = \boxed{\frac{15}{9}}$$

g. $y = x^2 - 6x + 5$ solve for x

$$1 \cdot x^2 - 6x + (5 - y) = 0$$

$a = 1$

$b = -6$

$c = 5 - y$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{6 \pm \sqrt{36 - 4(5-y)}}{2}$$

$$= \frac{6 \pm \sqrt{4(9-(5-y))}}{2}$$

$$= \frac{6 \pm 2\sqrt{9-(5-y)}}{2}$$

$$= 3 \pm \sqrt{9-(5-y)}$$

$9 - 5 + y$

$= 4 + y$

$$= \boxed{3 \pm \sqrt{4 + y}}$$